



**University of
Sunderland**

Faculty: BSc (Hons) Computer System Engineering

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MODULE CODE: CET140

MODULE TITLE: SPECIALIST PROJECT

MODULE LEADER: MANISHA KHATRI

SUBMISSION DATE AND TIME: 6/16/2026, 4 AM UK Time

ASSIGNMENT: 1

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1. Introduction:

Sleep is a biological need that directly affects physical, mental health and daily living. In spite of its importance poor sleep and unrest are increasingly common led by modern work demands, use of technology and also changing lifestyles.

Since, Artificial Intelligence is becoming more fast and this means that the machine learning techniques are becoming very useful for looking at sets of data and finding patterns that we might not see otherwise. Machine learning algorithms can see how different things are related to each other and use that to make guesses about what will happen next. When we talk about sleep Artificial Intelligence can help us look at things like how old someone's how stressed they are how much they exercise their heart rate, how long they sleep and other things about their health to figure out how well they sleep. Artificial Intelligence is really good at this because it can look at all these things together and make a prediction, about someone's sleep quality.

This project sits with BSc. Computer Science applying data science methods to real world health problems and aims to make meaningful and evidence-based findings.

2. Aims, Objectives and Problem Context

2.1 Problem Context:

Sleep illnesses is affecting many people globally and can cause to serious health problems which may include cardiovascular diseases, obesity and mainly depression. The dataset used in the project covers a wide range of behavioral and psychological variables like amount of caffeine consumption, screen time, stress, exercise, BMI and chronotype while making it ideal for identifying which factor strongly predict the poor sleep outcomes.

2.2 Aim:

The main aim of this project is to analyse the sleep health data using Python and identify the key daily lifestyle and physiological measures which helps to predict the sleep quality using Linear Regression as the main modeling method.

2.3 Objectives:

- To clean and preprocesses the 100,000 records dataset using Python Pandas.
- To handle the missing values, outliers and categorical variables.
- To conduct Exploratory Data Analysis (EDA) using Matplotlib and Seaborn to see the patterns and correlations between variables.
- To view the relationship between lifestyle factors and sleep outcomes.
- To Build a linear Regression model, Random Forest regression model, Decision Tree Regressor Model and using NumPy and Scikit-learn to predict the sleep_quality_score.
- To ensure that all work adheres to social, ethical, professional and legal standards.

3. Dataset Detail:

3.1 Overview and Source of Dataset:

The dataset contains 100,000 records and 32 column collected from individual across 15 countries

Table:

USA	20050	India	17840
Japan	8089	UK	8024
Brazil	7146	Germany	6026
Canada	4995	France	4982
Australia	4948	Italy	4018
South Korea	3971	Mexico	3888
Netherlands	3040	Spain	2011
Sweden	972		

Table : Number of Data from each country

It covers a very large range of demographic, behavioural, physiological, and sleep-specific variables, making it well-suited for analysis. The main target variable for regression is sleep_quality_score which is scored on a continuous scale from 1.0 to 10.0.

The source of the dataset is <https://www.kaggle.com/datasets/mohankrishnathalla/sleep-health-and-daily-performance-dataset>

3.2 Variable summary

Group	Columns	Purpose
Demographics	person_id, age, gender, occupation, bmi, country	Who the person is
Sleep Architecture	sleep_duration_hrs, sleep_quality_score, rem_percentage, deep_sleep_percentage, sleep_latency_mins, wake_episodes_per_night	How they slept
Lifestyle	caffeine_mg_before_bed, alcohol_units_before_bed, screen_time_before_bed_mins, exercise_day, steps_that_day, nap_duration_mins	What they did
Psychological	stress_score, work_hours_that_day, chronotype, mental_health_condition	How they felt

Group	Columns	Purpose
Health Context	heart_rate_resting_bpm, sleep_aid_used, shift_work, room_temperature_celsius, weekend_sleep_diff_hrs	Health modifiers
Environmental	season, day_type	When the record is from
Sleep Outcome Variables	cognitive_performance_score, sleep_disorder_risk, felt_rested	Help in prediction

4. Literature Review:

Tanuku and Tummala matched many models like linear Regression , Ridge Regression, Lasso Regression and Random Forest Regressor by using biological factors such as heart rate, respiration rate, body temperature and blood oxygen level. Among all the model used Linear Regression and Ridge Regression got the best result with and R^2 score of 99% and RMSE of 0.02 which outperformed Random Forest (98%) and Lasso Regression (95%). This study showed the efficiency of regression models for sleep quality prediction but it was limited because of the use of single dataset and lack of external validation (Tanuku, 2023)

(Alazaidah, 2024) calculated Linear Regression, SVM, Random Forest and Artificial Neural Networks to predict sleep related tasks by using behavioral and physiological data. Ensemble based models achieved more accuracies which is above 90% which outperformed old statistical methods. But problems like missing data, class imbalance and model interpretability has being the major limitations (Alazaidah, 2024).

(Hamza, et al., 2023) advised a Deep Belief Network(DBN) along with Sand Grouse Optimization(SGO) by using a Kaggle sleep dataset which contained 400 records classified into four sleep quality categories. There WSHMSQP-ODL model got a classification accuracy of 97.5% which outperformed Long Short Term Memory (LSTM) models 91.7% and other benchmarks algorithms which shows the impact of deep learning and optimization techniques for sleep quality prediction. But this study was limited to its relatively small dataset size and dependence on a single data source which affected the models. (Hamza, et al., 2023)

(Halsen, et al., 2022) used multiple linear regression on actigraphy from 203 elite athletes (1,975 nights). They explained **78% of variance in total sleep time** ($R^2=0.78$, $RMSE \approx 21$ min) but only **22% in sleep efficiency** ($R^2=0.22$, $RMSE \approx 6.5\%$). Key predictors were bedtime/offset times.

4.1 Research Gap:

While the ensembles and deep learning techniques provide high accuracy results, not many studies have explored the use of simple linear regression models incorporating several sleep features.

5. Methodology:

5.1 Approach:

It is the project where it uses quantitative, data driven approach. Dataset will be analyzed in Python inside a Jupyter Notebook for reproducibility. The methodology will follow four phases i.e preprocessing, EDA, Statistical analysis and regression modeling.

5.2 Data Preprocessing using Pandas and NumPy:

- Load CSV file using `pd.read_csv()` and look for the shape, dtypes and null values.
- Handling the null or missing values in the datasets
- Determine categorical columns like gender, occupation, chronotype, season, mental_health_condition using label encoding or one hot encoding.
- Data normalization where necessary

5.3 Exploratory Data Analysis (EDA) using Matplotlib and Seaborn:

- Histograms will be made for analysis
- Boxplot to compare sleep quality across the occupation, gender , chronotype groups.
- Correlation heatmaps will be made for all numerical values
- Scatter plots of key predictor pairs.

5.4 Model Development

A Linear Regression model was implemented using the Scikit-learn library. The dataset was divided into training and testing sets the dataset is split 80% training / 20% testing. The model learned the relationship between independent variables and sleep quality scores. Model performance is measured using R^2 (how much of the variation in sleep quality the model explains) and RMSE (the average size of prediction errors). Residual plots check that the model assumptions are satisfied.

5.5 Tools Used:

Tools	Purpose
Pandas	For Data loading, cleaning and transformation
NumPy	For Numeric operation, normalization and Regression Calculation
Matplotlib	For histograms, scatter plots and residual plots
Seaborn	For Heatmaps, boxplot and pair plots
Scikit-learn	For linear Regression model, train/test and split, R^2 evaluation
Jupyter Notebook	For reproducible and documented analysis environment

6. Social, Ethical, Professional and Legal Issues (SEPLi):

6.1 Social Issues:

Since Sleep health data have variables like `meantal_health_condition`, `occupation` and `shift_work` these holds a social responsibility. If the analysis shows that some groups like nurses, drivers or shift worker have lower sleep quality scores then these results must not be used to showcase those individuals. These results only reflects the systemic working conditions rather than personal failings. Also the dataset have 15 countries with different culture towards the sleep,work and mental health. Before any conclusion these topics must be acknowledged and also acknowledge the cultural diversity.

6.2 Ethical Issues:

Since the dataset used in this project is available publicly and have full details of the author and is fully anonymized which means no individual can be or is identified from the records. But stil the ethical responsibility remains. The analysis is fully correlational columns like `stress_score` or `caffeine_mg_before_bed` may be connected to the poor sleep quality but this doesn't prove they cause it and no casual claims will be made in the reporting of the result. Also, linear regression model gives the predicted sleep quality scores that are for academic purposes only and never be used as a clinical diagnostic or medical advice.

6.3 Professional Issues:

As a computer professional this project follows to British Computer Society (BCS) Code of Conduct where there should be honesty, integrity and competence in the work. This means truly reporting both the weakness and strengths of the model used no just presenting results that looks satisfactory. All Python code used during the project will be fully explained and commented, logically structured and stored using version control to make sure that it is reproducible and auditable. All the libraries used in the project like Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn will be acknowledged.

6.4 Legal Issues:

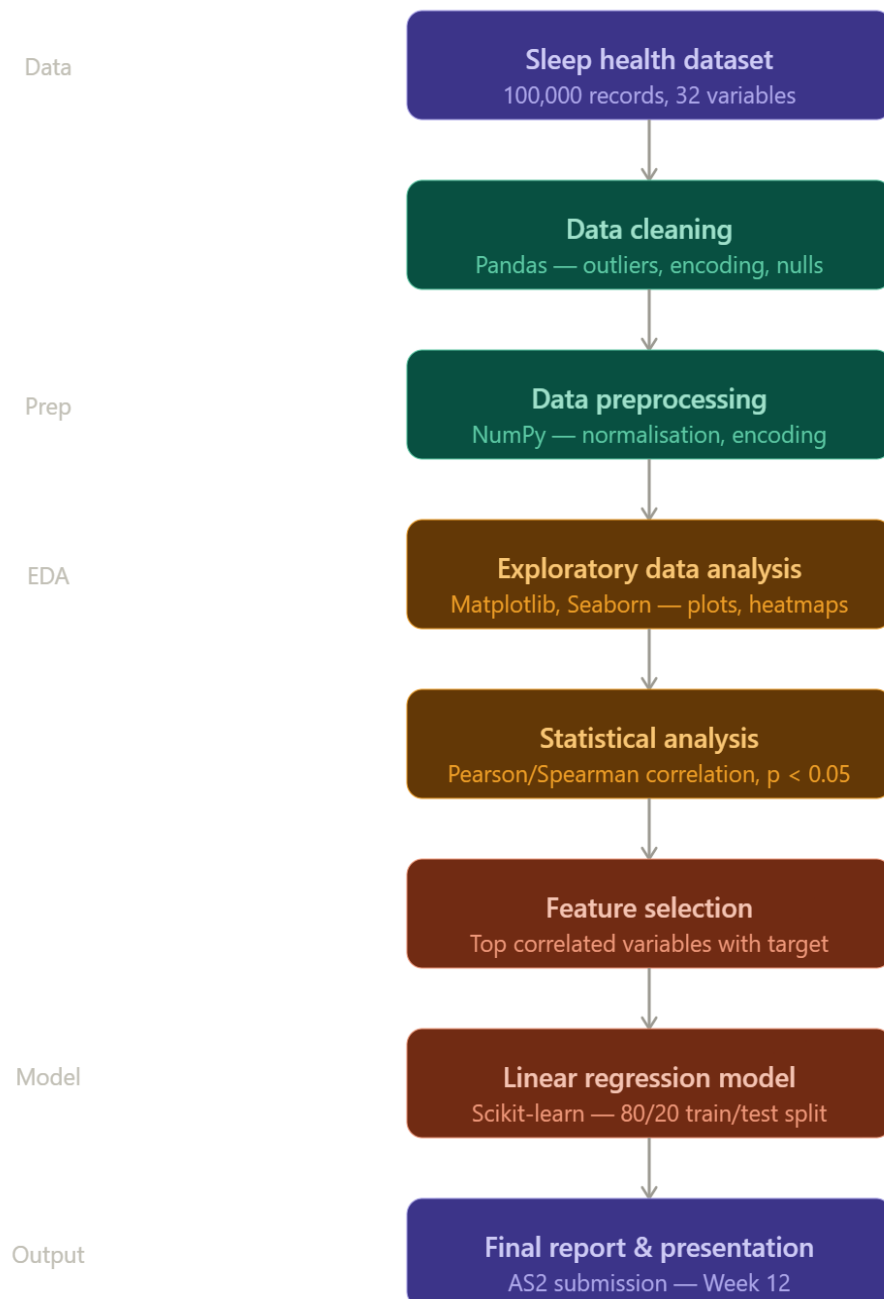
Even the dataset doesn't contain any personal information and therefore falls outside the strict duties of the UK General Data Protection Regulation (UK GDPR) so this project will treat the data as confidential and according to the GDPR principles. The principle of data minimization will be followed by using only column which are related to the research questions and also the principle od purpose limitation will be respected by making sure that the data is only use by only me and not shared. The Computer Misuse Act 1990 is also applicable as a professional standard and all data access and processing will be according to the law, transparently and kept to the tools and environment approved for this project.

7. Project Plan

7.1 Overview:

The project will be run for 12 weeks from research to final submission. This project will focus in planning, research, data analysis and model development. Each task is given time to as specific week to make sure the constant progress and timely completion.

Project Flowchart:



7.2 Gantt Chart:

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Project Planning												
Proposal												
Literature Review & Research Gaps												
Data Collection & Cleaning												
Data Preprocessing												
Exploratory Data Analysis (EDA)												
Model Development												
Model Evaluation												
Draft Report												
SEPLi + Conclusion												
Assignment 1 Submission												
Final Report & Presentation Assignment 2 submission												

7.3 Week Vs Task Table:

Week 1	Project planning and scope defined	Project brief
Week 2	Proposal drafted and submitted	Proposal doc
Week 3	Literature review and research gaps completed	Reading notes
Week 4	Dataset collected and cleaned	Clean dataset
Week 5	Data preprocessing complete	Processed data
Week 6	Exploratory Data Analysis finished	EDA Notebook
Week 7	Linear Regression model developed	Model notebook
Week 8	Model evaluated and results interpreted	Model results
Week 9	Draft report written	Draft report
Week 10	SEPLi and conclusion completed	SEPLi section
Week 11	Assignment 1 submitted	AS1 Submission
Week 12	Final report and presentation	AS2 + Present

References

Alazaidah, R. A. M. A.-K. M. a. A. S., 2024. Machine learning approaches for sleep disorder prediction and diagnosis: A systematic review. *Diagnostics*, 14(1), p. 27.

Halsen, S. L. et al., 2022. Sleep Regularity and Predictors of Sleep Efficiency and Sleep Duration in Elite Team Sport Athletes. *Sports Medicine*, 8(1), p. 79.

Hamza, M. A. et al., 2023. ‘Wearables-Assisted Smart Health Monitoring for Sleep Quality Prediction Using Optimal Deep Learning. *Sustainability*, 15(2), p. 1084.

Tanuku, P. a. T. S., 2023. Integration of Feature Selection Techniques Using a Sleep Quality Dataset for Comparing Regression Algorithms.